

Ayush Tiwari

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EDUCATION

City Montessori School, Lucknow — Class XII

2022

SRM University Delhi-NCR, Sonepat, B.Tech in Computer Science Engineering

2023 – 2027

SUMMARY

Backend engineer specializing in distributed systems, high-concurrency services, and real-time architectures. Experienced in building fault-tolerant systems, scalable APIs, and network-driven applications using Go. Focused on low-latency infrastructure, observability, and resilient backend design.

TECHNICAL SKILLS

Languages	Go, C, C++, TypeScript, Python, Bash
Backend	Fastify, Express, REST APIs, WebSockets
Infrastructure	Docker, Kubernetes, AWS, Terraform, GitHub Actions, Nginx
Observability	Grafana, Loki, Prometheus
Databases	PostgreSQL, Redis, MongoDB
Systems	Linux, Distributed Systems, TCP/UDP, Microservices

KEY PROJECTS

Cerver | C, JavaScript, Compiler

[@velox0/Cerver](#)

- Built a compile-time web framework transpiling restricted JS into optimized standalone C binaries with 0 node dependencies.
- Implemented Next.js-style file-based routing generating native C code with epoll/kqueue for consistent sub-5ms response times.
- Added automatic asset embedding & minification, producing <900KB single binaries with <2ms startup time on M4 MacBook.
- Integrated an npm-based CI/CD pipeline for automated builds, testing, and package publishing to the npm registry.
- Achieved exceptional performance of 125k+ req/s with sub-10ms p99 latency under heavy load on MacBook Air M4.

Kraken | Go, PostgreSQL, Redis, Docker, SPA

[GitHub/Kraken](#)

- Architected distributed uptime monitoring system maintaining 99.95% uptime with incident deduplication across endpoints.
- Engineered automated remediation engine executing custom shell scripts with rate limiting on detected failures.
- Implemented strict sandboxed script execution engine with environment abstraction and zero known attack surface.
- Designed flexible architecture scaling seamlessly from single monolithic binary to Redis-backed microservices.
- Built secure SPA dashboard with custom RBAC system and brute-force protection for reliable incident visibility.

Moonlight-Server | Go, Distributed Systems

moonlight.velox0.com

- Developed lightweight high-concurrency WebSocket proxy handling 3,000+ simultaneous connections on small EC2 instances.
- Implemented region-aware load balancing prioritizing nodes dynamically based on connection recency and latency metrics.
- Configured robust access token security alongside comprehensive real-time observability and health metrics dashboard.
- Designed hierarchical region fallback mechanism ensuring continuity by routing to global nodes during local outages.
- Engineered deterministic task tracking with automatic WebSocket-to-HTTP protocol degradation for seamless reliability.

Gatekeeper | Go, Reverse Proxy, Authentication

[GitHub/Gatekeeper](#)

- Developed lightweight virtual-host-aware reverse proxy with multi-port listeners handling 5k+ req/s on constrained hardware.
- Implemented hierarchical configuration system with zero-downtime reloads and graceful shutdown for production stability.
- Built customizable animated login plugins (aurora, matrix, hearts) with per-host theming and secure session management.
- Designed bcrypt-based authentication with configurable TTL supporting consistent daily active user sessions.
- Added interactive CLI tool along with comment-preserving config handling for streamlined enterprise deployment.

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- Developed and deployed a personal portfolio platform showcasing systems, backend, projects with GitHub integration.
- Migrated backend architecture from Express.js to Fastify, improving performance, and request handling efficiency.
- Provisioned and managed AWS EC2 infrastructure using Terraform, including IAM configuration, deployment automation.
- Built automated CI/CD pipelines with GitHub Actions for testing, building, and deployment workflows across multiple projects.
- Integrated live project, commit, and activity feeds from GitHub APIs to surface real time development updates.

ACHIEVEMENTS & ADDITIONAL EXPERIENCE

- Smart India Hackathon (SIH) 2024 Finalist:** Developed a geospatial AI pipeline utilizing SAR data for real-time human-induced change detection.
- Systems Management:** Expertise in administrating complex Linux environments, including automated local-to-cloud synchronization via custom network tooling.